
Yichong Zhang

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[Personal website](#)

Education

Tsinghua University, Weiyang College, Beijing, China Aug 2023 – Jul 2027 (expected)

- Undergraduate in Mathematics and Physics + Electrical Engineering, GPA: 3.9/4.0, Rank: 3/52

Harvard College, Harvard University, Cambridge, MA, United States Aug 2025 – May 2026

- Visiting Undergraduate Student (VUS)

Research Interests

My research focuses on building efficient and reliable AI systems through cross-stack optimization and automated systems development. My current interests include:

- **Systems and architecture for emerging AI workloads**, particularly efficient real-world deployment of embodied AI.
- **Reliable code generation and optimization with coding agents**, with a focus on testing and verification.

Publications

Yichong Zhang, Genghan Zhang, Zishen Wan, Kunle Olukotun, Olivia Hsu, “*Behavioral Test Diversity: Quantifying Test Suite Diversity Through Agentic Code Repair*,” submitted to ICLR 2027

Jiayi Qian, **Yichong Zhang**, Hanchen Yang, Chun Tao, Souvik Kundu, Zishen Wan, Tushar Krishna, “*Serving Agentic Workflows with a Physical-Plan Compiler and Adaptive Runtime*,” submitted to ASPLOS 2027

Ke Hong, Xiuhong Li, Minxu Liu, Qiuli Mao, Tianqi Wu, Zixiao Huang, Lufang Chen, Zhong Wang, **Yichong Zhang**, Zhenhua Zhu, Guohao Dai, Yu Wang, “*Efficient and Adaptable Overlapping for Computation and Communication via Signaling and Reordering*,” accepted by EuroSys 2026

Tianchen Zhao, Ke Hong, Xinhao Yang, Xuefeng Xiao, Huixia Li, Feng Ling, Ruiqi Xie, Siqi Chen, Hongyu Zhu, **Yichong Zhang**, Yu Wang, “*PAROAttention: Pattern-Aware ReOrdering for Efficient Sparse and Quantized Attention in Visual Generation Models*,” accepted by NeurIPS 2025

Research Experience

Tests as Specifications for Reliable Code Generation May 2026 – Present

Advisors: Profs. Kunle Olukotun & Olivia Hsu, Department of Computer Science, Stanford University

- Investigated how effectively tests specify intended program behavior for coding agents, identifying limitations of existing test-suite metrics in measuring the diversity of behavioral feedback.
- Developed Behavioral Test Diversity (BTD), which quantifies test-suite diversity by measuring whether code repairs for one test satisfy other tests.
- Applied BTD to guide test-suite pruning and expansion, reducing tests supplied as executable specifications by 61–90% across ProgramBench, vLLM, and ComputeEval while preserving full test suite recovery, and achieving 3.2× higher behavioral-diversity gain than a flat expansion baseline.

Benchmarking and Characterizing Robotic Manipulation Workloads Feb 2026 – Present

Advisors: Profs. Zishen Wan & Vijay Janapa Reddi, Harvard SEAS, Harvard University

- Developed EmbodiedPerf, a lightweight profiling framework for robotic manipulation benchmarks, enabling measurements of latency, energy consumption, and hardware utilization alongside task success.
- Benchmarked robotic manipulation policies (VLA, WAM, and Code-as-Policy) using EmbodiedPerf across LIBERO, RoboTwin, and BEHAVIOR-1K on H100, RTX 5090, and Jetson Thor and Orin, revealing that smaller model sizes do not necessarily translate into better task-level efficiency and that performance bottlenecks vary across hardware.

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- Evaluated different power-management configurations (including power-budget modes and DVFS) on Jetson platforms and asynchronous inference across models, finding that inference-level speedups do not necessarily translate into task-level efficiency gains.
 - Deployed and evaluated the profiling framework on a Franka robot arm performing a real-world pick-and-place task, with $\pi 0.5$ inference running on an RTX 5090 GPU.

Efficiency and Robustness of Vision-Language-Action Models

Sep 2025 – Feb 2026

Advisors: Profs. Vijay Janapa Reddi & Zishen Wan, Harvard SEAS, Harvard University

- Evaluated cache reuse and 4- and 8-bit weight-only quantization in VLA models, observing no apparent degradation in task performance or robustness under the tested optimization settings.
- Characterized aggressive low-bit quantization in OpenVLA and OpenVLA-OFT language backbones, observing minimal performance degradation at 3 bits (0.9%) but substantial degradation at 2 bits (16%) on LIBERO.
- Implemented and compared SVD- and RRQR-based low-rank approximations of VLA language backbones, finding that SVD consistently outperformed RRQR across the evaluated ranks; further analyzed relationships between dominant weight subspaces and action primitives.

GPU Systems and Kernel Optimization

Jan 2025 – Aug 2025

Advisor: Prof. Yu Wang, Department of Electronic Engineering, Tsinghua University

- Implemented fused CUDA reorder kernels for LayerNorm and RoPE in a hardware–software co-designed sparse-attention system for video-generation transformers, achieving $\sim 20\times$ speedup over PyTorch implementations.
- Built end-to-end benchmarking pipelines and test benches for PAROAttention, and conducted kernel-level performance comparisons against FlashAttention-2, XAttention and SVG.
- Contributed to a signaling-based computation–communication overlap mechanism for GPU workloads and benchmarked GEMM–collective workloads, demonstrating up to $1.65\times$ speedup over non-overlapped execution on RTX 4090 GPUs.

Extracurricular Activities

"A GreatPerhaps" Career Talk Project Beijing, China

Mar 2024 – Jan 2025

Founder

- Founded and led a cross-university team spanning Tsinghua, Peking, Shanghai Jiao Tong, Fudan, and the University of Washington, producing career-focused interviews and content with $\sim 100,000$ cumulative views.
- Conducted one-on-one interviews with senior technology executives, including CTOs from Megvii Technology, Pudu Robotics, and TeleAI.
- Project website: <https://grephap.org>

Academic Service

Reviewer: ICLR 2027, NeurIPS 2026 Workshop (Verify-Agents Workshop)

Skills

Language: Chinese (Native); English (Fluent, with a TOEFL score of 106)

Programming and Software: Python, CUDA, C++, Bash, Git, LaTeX